

Taller de Ingeniería Informática - Laboratorio N2

Grupo X - Integrantes: Nombre1, Nombre2, Nombre3, Nombre4

June 3, 2026

1 Introducción

Este informe documenta el despliegue de una aplicación web dinámica con operaciones CRUD utilizando contenedores Docker, orquestados con docker-compose, una base de datos MySQL y persistencia mediante volúmenes.

2 Desarrollo

2.1 Estructura de la aplicación

Se implementó una aplicación en PHP con MySQL que permite Crear, Leer, Actualizar y Eliminar registros de una tabla "items". La arquitectura se compone de dos contenedores:

- **web:** Contenedor PHP con Apache, que expone el puerto 8080.
- **db:** Contenedor MySQL 8.0 con persistencia en volumen.

Ambos se comunican a través de una red Docker interna llamada `crud_network`.

2.2 Evidencias de ejecución

3 Conclusión

Se logró desplegar exitosamente una aplicación web CRUD utilizando Docker, cumpliendo todos los requisitos: contenedores separados, volumen persis-

```
root@debian:~# cd /root/practico2
root@debian:~/practico2# docker compose build
[+] Building 4.1s (10/10) FINISHED
=> [internal] load local bake definitions
=> reading from stdin 452B
=> [internal] load build definition from dockerfile
=> transferring dockerfile: 138B
=> [internal] load metadata for docker.io/library/php:8.2-apache
=> [internal] load .dockerignore
=> transferring context: 2B
=> [1/3] FROM docker.io/library/php:8.2-apache@sha256:a1fcc431bd3acaa9a6394a71d162726fc0a6e4bea0400a3b94f925b6130658dd
=> resolve docker.io/library/php:8.2-apache@sha256:a1fcc431bd3acaa9a6394a71d162726fc0a6e4bea0400a3b94f925b6130658dd
=> [internal] load build context
=> transferring context: 7.35kB
=> CACHED [2/3] RUN docker-php-ext-install pdo pdo_mysql
=> [3/3] COPY . /var/www/html/
=> exporting to image
=> exporting layers
=> exporting manifest sha256:1460025cd91614ed57f52f1ab92f3eb6fb26c9c188021a95ee91b913cebd13
=> exporting config sha256:e3cf6414ab7c515e3d3acadd12ed325dbd7fb3af4e7dbd0db8f2fa92a1b04f
=> exporting attestation manifest sha256:f946add71ef2eb00dfc36d9db0a1b78f91c9331403bcb093d4d8953e88fb74
=> exporting manifest list sha256:4d99b0354007709d9e8edf45e979abccc04135050b50424ccc0771db7842f019
=> naming to docker.io/library/practico2-web:latest
=> unpacking to docker.io/library/practico2-web:latest
=> resolving provenance for metadata file
[+] build 1/1
Image practico2-web Built
root@debian:~/practico2#
```

Figure 1: Ejecución de docker-compose build

```
root@debian:~/practico2# docker compose up -d
[+] up 2/2
Container mysql_crud_db Running
Container php_crud_app Started
root@debian:~/practico2# docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED    STATUS      PORTS                               NAMES
c9b9baaa9012   practico2-web  "docker-php-entrypoi..." 54 seconds ago Up 54 seconds 0.0.0.0:8080->80/tcp, [::]:8080->80/tcp   practico2-web
07a1f5c580a   mysql:8.0     "docker-entrypoint.s..." 51 minutes ago Up 2 minutes 0.0.0.0:3306->3306/tcp, [::]:3306->3306/tcp, 3306/tcp   mysql_crud_db
da55c035064   odoo:18       "/entrypoint.sh odoo..." 31 hours ago Up 2 minutes 0.0.0.0:8069->8069/tcp, [::]:8069->8069/tcp, 8071-8072/tcp   odoo-18
7f38d9d5d1de   postgres:15   "docker-entrypoint.s..." 31 hours ago Up 2 minutes 5432/tcp   postgres
root@debian:~/practico2#
```

Figure 2: docker-compose up -d y docker ps

```
2026-06-03 20:12:36+00:00 [Note] [Entrypoint]: Stopping temporary server
2026-06-03T20:12:36.402602Z 14 [System] [MY-013172] [Server] Received SHUTDOWN from user root. Shutting down mysqld (Version: 8.0.46).
2026-06-03T20:12:38.295335Z 0 [System] [MY-019310] [Server] /usr/sbin/mysqld: Shutdown complete (mysqld 8.0.46) MySQL Community Server - GPL.
2026-06-03 20:12:38+00:00 [Note] [Entrypoint]: Temporary server stopped

2026-06-03 20:12:38+00:00 [Note] [Entrypoint]: MySQL Init process done. Ready for start up.

2026-06-03T20:12:38.759152Z 0 [Warning] [MY-011068] [Server] The syntax '--skip-host-cache' is deprecated and will be removed in a future release. Please use 85
r GLOBAL host_cache_size=0 instead.
2026-06-03T20:12:38.760475Z 0 [System] [MY-019110] [Server] /usr/sbin/mysqld (mysqld 8.0.46) starting as process 1
2026-06-03T20:12:38.778652Z 1 [System] [MY-013376] [InnoDB] InnoDB initialization has started.
2026-06-03T20:12:39.512735Z 1 [System] [MY-013377] [InnoDB] InnoDB initialization has ended.
2026-06-03T20:12:39.696382Z 0 [Warning] [MY-010669] [Server] CA certificate ca.pem is self signed.
2026-06-03T20:12:39.740230Z 0 [System] [MY-015892] [Server] Channel mysql_main configured to support TLS. Encrypted connections are now supported for this chan
el.
2026-06-03T20:12:39.916462Z 0 [Warning] [MY-011819] [Server] Insecure configuration for --pid-file: Location '/var/run/mysqld' in the path is accessible to all
os users. Consider choosing a different directory.
2026-06-03T20:12:39.948950Z 0 [System] [MY-011323] [Server] X Plugin ready for connections. Bind-address: '::' port: 33060, socket: /var/run/mysqld/mysqld.sock
2026-06-03T20:12:39.955222Z 0 [System] [MY-019311] [Server] /usr/sbin/mysqld: ready for connections. Version: '8.0.46' socket: '/var/run/mysqld/mysqld.sock' p
ort: 3306 MySQL Community Server - GPL.
2026-06-03 21:00:47+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 8.0.46-1.el9 started.
2026-06-03 21:00:48+00:00 [Note] [Entrypoint]: Switching to dedicated user 'mysql'
2026-06-03 21:00:48+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 8.0.46-1.el9 started.
/var/lib/mysql/mysql.sock -> /var/run/mysqld/mysqld.sock
2026-06-03T21:00:50.054700Z 0 [Warning] [MY-011068] [Server] The syntax '--skip-host-cache' is deprecated and will be removed in a future release. Please use 85
r GLOBAL host_cache_size=0 instead.
2026-06-03T21:00:50.726942Z 0 [System] [MY-019110] [Server] /usr/sbin/mysqld (mysqld 8.0.46) starting as process 1
2026-06-03T21:00:50.750173Z 1 [System] [MY-013376] [InnoDB] InnoDB initialization has started.
2026-06-03T21:00:52.407708Z 1 [System] [MY-013377] [InnoDB] InnoDB initialization has ended.
2026-06-03T21:00:52.592754Z 0 [System] [MY-010223] [Server] Starting XA crash recovery...
2026-06-03T21:00:53.024142Z 0 [System] [MY-010223] [Server] XA crash recovery finished.
2026-06-03T21:00:53.417422Z 0 [Warning] [MY-010669] [Server] CA certificate ca.pem is self signed.
2026-06-03T21:00:53.417152Z 0 [System] [MY-015892] [Server] Channel mysql_main configured to support TLS. Encrypted connections are now supported for this chan
el.
2026-06-03T21:00:53.437322Z 0 [Warning] [MY-011819] [Server] Insecure configuration for --pid-file: Location '/var/run/mysqld' in the path is accessible to all
os users. Consider choosing a different directory.
2026-06-03T21:00:53.565952Z 0 [System] [MY-011323] [Server] X Plugin ready for connections. Bind-address: '::' port: 33060, socket: /var/run/mysqld/mysqld.sock
2026-06-03T21:00:53.567890Z 0 [System] [MY-019311] [Server] /usr/sbin/mysqld: ready for connections. Version: '8.0.46' socket: '/var/run/mysqld/mysqld.sock' p
ort: 3306 MySQL Community Server - GPL.
mysqladmin: [Warning] Using a password on the command line interface can be insecure.
```

Figure 3: Logs del contenedor MySQL mostrando inicio correcto

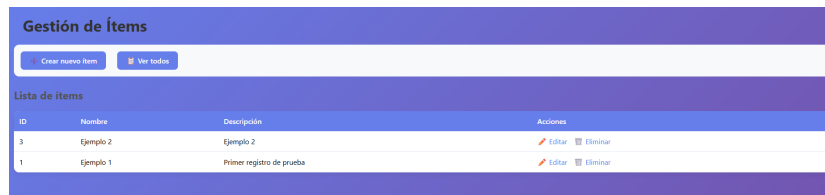


Figure 4: Aplicación corriendo en `http://localhost:8080` con operaciones CRUD funcionales

tente, red virtual, y documentación completa. El sistema es reproducible y portable.

4 Anexos

- Repositorio GitHub: <https://github.com/Vicho255/Practico-2.git>